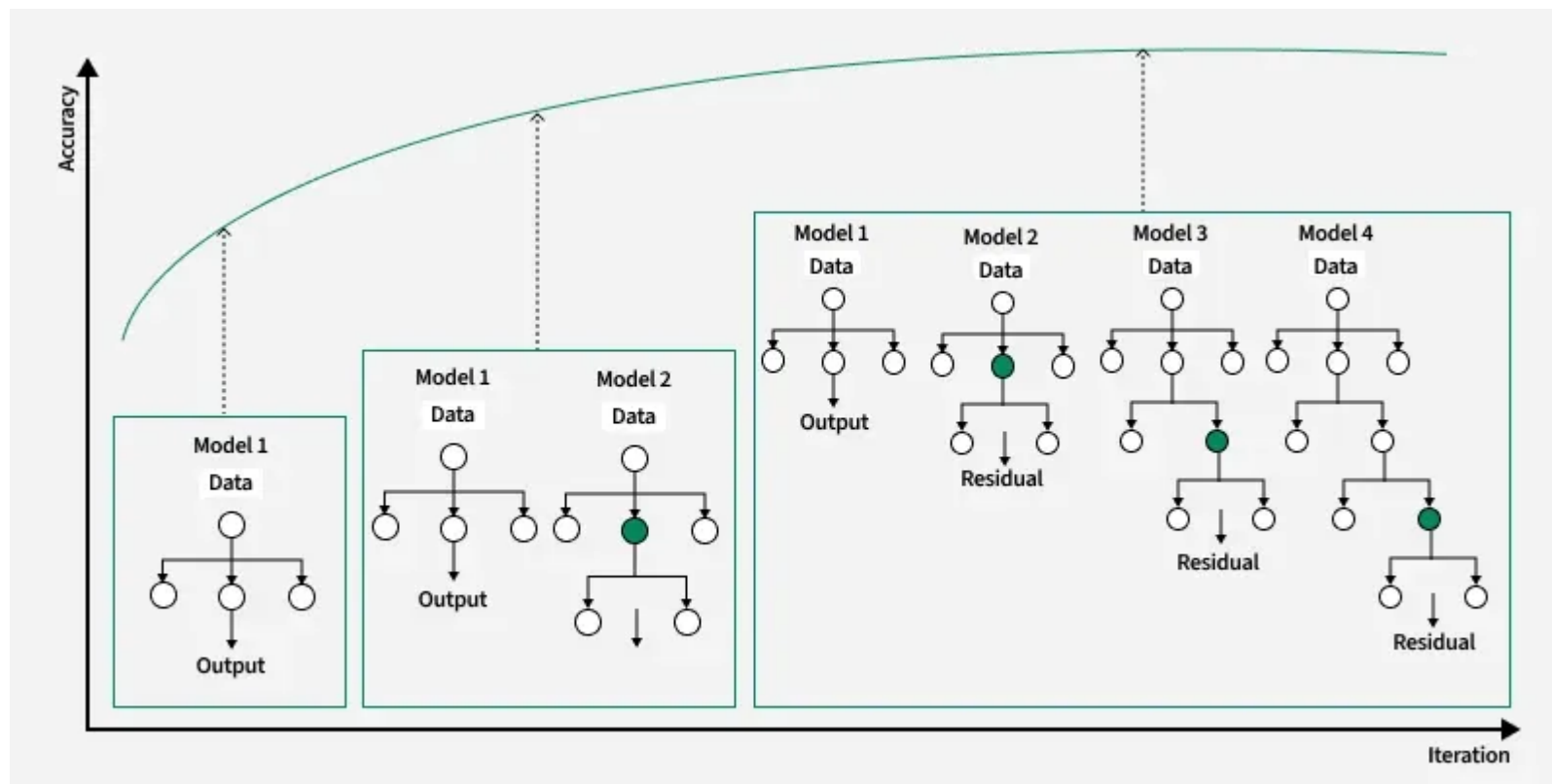


LG BOOST REGRESSOR



Gradient Boosting

- ▶ Gradient Boosting is an ensemble machine learning technique that builds a series of decision trees.
- ▶ Each decision trees aimed at correcting the errors of the previous ones.
- ▶ Unlike AdaBoost, which uses shallow trees, Gradient Boosting uses deeper trees as its weak learners.
- ▶ Each new tree focuses on minimizing the residual errors – the differences between actual and predicted values – rather than learning directly from the original targets.

PROCESS

- ▶ Starts with an initial baseline prediction - the average value of the target
- ▶ Calculates the *residuals* of the predictions.
- ▶ Trains a new model specifically to predict these residuals.
- ▶ Adds the new model's predictions to the previous ones to reduce the overall error.

Thank You!